FI FISHINTEL GLOBAL

ENVIRONMENTAL RECONSTRUCTION REPORT

Decision-Time Environmental Reconstruction

Anchorage Failure During Forecast Storm Conditions

INCIDENT WINDOW	Winter storm period
REPORT REFERENCE	FIG-FLAGSHIP-001
VESSEL CATEGORY	Bulk carrier, in ballast
REPORT VERSION	v2.0
OPERATIONAL CONTEXT	UK coastal lee shore anchorage
CLASSIFICATION	Public — Anonymised

ENVIRONMENTAL RECONSTRUCTION COMPLETE — ALL DATA VERIFIED

QUALIFIED

This report presents a complete environmental reconstruction from ERA5 reanalysis data. The composite risk score incorporates two supplementary parameters (atmospheric pressure, air temperature) at standard reference values rather than case-specific measurements. All primary risk drivers — wind speed, gust speed, and significant wave height — are derived entirely from ERA5.

Protocol v1.1 Report v2.0 Renderer v2.0.0

Report ID: FIG-FLAGSHIP-001

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Executive Summary

Incident type: Grounding following anchor drag. A foreign-registered bulk carrier, anchored in ballast at a UK coastal anchorage, dragged its anchor during a severe winter storm and grounded on a public beach. Successfully refloated. No injuries, no pollution.

Core question: What environmental conditions were objectively knowable at the time the decision was made to remain at anchor on a lee shore overnight, and at what point did those conditions make the vessel's position untenable?

Method: ERA5 atmospheric reanalysis data was extracted at the anchorage coordinates across a five-point time window (21:00–05:00 UTC) and processed through the FishIntel deterministic reconstruction pipeline.

Key result: The vessel was in an ELEVATED-risk environmental regime from 21:00 UTC, escalating to EXTREME risk by 00:00 UTC — five hours before the grounding. All three monitored environmental signals (wind, gust, wave) exceeded safety thresholds simultaneously at 21:00 UTC, providing an 8-hour intervention window.

Analytical finding: Environmental reconstruction shows conditions were deteriorating gradually and predictably across the entire incident window. The narrative description of Force 9–10 appears to conflate locally generated wind with remotely generated swell. ERA5 shows sustained wind of 20–27 knots (Beaufort force 5–7), not the 47–55 knots implied by the narrative Beaufort classification.

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Incident Overview

Vessel Profile

TYPE	Bulk carrier
FLAG	Foreign-registered
LENGTH OVERALL	~180 m
GROSS TONNAGE	~16,000
YEAR OF BUILD	Early 1980s
CARGO CONDITION	In ballast
PROPULSION	Direct drive, single fixed-pitch propeller, no bow thruster
CREW	18
ANCHORS	Two, similar to AC14 type. Port: 11 shackles. Starboard: 10 shackles.
P&I INSURANCE	Not in force at time of incident

The vessel had sailed from a European port following a lengthy port state control detention. Unable to take bunkers before departure due to unpaid debts, she was ordered to proceed to North Africa for refuelling but had insufficient fuel for the passage.

She anchored in the bay approximately 1 nautical mile off a public beach, in a charted depth of 17 metres, using the port anchor with 7 shackles of cable.

On the evening before the grounding, the master was aware that the weather forecast predicted southerly winds up to Beaufort force 8. He wrote in his night orders that the main engine should be at immediate readiness for wind speeds exceeding 16–17 m/s.

By 0400 the following morning, the wind was gusting over 52 knots with a 4–5 metre swell. At 0425, the anchor began to drag. Despite full engine power and helm, the master was unable to turn the vessel into the wind. She grounded at 0525.

The vessel was refloated by harbour tugs at 1415 the same day. An underwater survey revealed hull plate deformation, breached tanks, and partial bilge keel detachment. There were no injuries and no pollution.

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Environmental Escalation Timeline

ERA5 reanalysis data was extracted at the incident coordinates across 5 timestamps. The following table presents reconstructed conditions and composite risk score at each extraction point.

TIME (UTC)	EVENT	SUSTAINED WIND	GUSTS	SIG. WAVE HT	RISK SCORE	BAND
21:00	*Initial assessment*	20.1 kt	32.0 kt	2.7 m	60.1%	ELEVATED
00:00	*Escalation to EXTREME*	23.5 kt	37.9 kt	3.3 m	91.4%	EXTREME
03:00	*Conditions continuing*	25.3 kt	41.6 kt	3.8 m	100.0%	EXTREME
04:00	*Conditions continuing*	26.2 kt	43.7 kt	4.0 m	100.0%	EXTREME
05:00	*Lead timestamp*	27.0 kt	44.8 kt	4.1 m	100.0%	EXTREME

SCORING METHODOLOGY

Risk classifications are derived from a domain-calibrated model developed against historical maritime environmental conditions. The model assigns ordinal severity bands (LOW through EXTREME) based on the combined effect of wind, gust, wave, and contextual factors. The model has not been independently peer-reviewed. The environmental reconstruction data underlying the scores is independently verifiable from ERA5.

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Narrative Weather Description

The following environmental conditions were described in the original accident investigation narrative.

PARAMETER	NARRATIVE DESCRIPTION	OBSERVATIONS
Wind (sustained)	Southerly, Beaufort force 9–10 Equivalent to 47–55 knots	*Described at time of incident. No hourly progression provided.*
Wind (gusts)	Gusting over 52 knots Reported by 0400	*Single threshold. No peak gust value stated.*
Wind direction	Southerly	*Cardinal direction only. No bearing given.*
Sea / swell	4–5 m swell	*Range given. No wave period or direction stated.*
Forecast (evening before)	Southerly up to Beaufort force 8 34–40 knots	*Forecast available to master. Conditions exceeded this forecast.*
Atmospheric pressure	Not stated	*No barometric data in narrative account.*
Air temperature	Not stated	—
Visibility	Not stated	—
Tidal stream	Not stated	*Not described in the investigation narrative.*

PARAMETER	NARRATIVE DESCRIPTION	OBSERVATIONS
Darkness	Incident occurred in darkness (0425–0525 UTC, December)	*Confirmed by timestamp.*

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Environmental Reconstruction

The following table compares narrative conditions with ERA5 reconstruction values. All reconstructed values are produced by the deterministic FishIntel pipeline.

PARAMETER	NARRATIVE	FISHINTEL RECONSTRUCTION	ASSESSMENT
Wind speed (sustained)	Beaufort F9–10 (47–55 kt)	27.0 kt	*Lower than narrative Beaufort estimate. Likely reflects separation between local 10 m wind and swell-driven sea-state severity.*
Wind gusts	Gusting over 52 knots	44.8 kt	*Broadly consistent with severe gust loading. ERA5 gust at 10 m is within ~15% of bridge-height observation.*
Wind direction	Southerly	179°	*Confirmed due south. Consistent with lee shore exposure at anchorage.*
Significant wave height	4–5 m swell	4.1 m	*Strong alignment. Validates ERA5 capture of the storm event.*

PARAMETER	NARRATIVE	FISHINTEL RECONSTRUCTION	ASSESSMENT
Atmospheric pressure	Not stated	1004.4 hPa	*Below-average pressure consistent with active low.*
Air temperature	Not stated	11.8 °C	*Mild for winter; consistent with maritime air mass from the south.*

27.0 kt	44.8 kt	4.1 m	1004.4 hPa
SUSTAINED WIND (10 M)	GUST SPEED	SIG. WAVE HEIGHT	PRESSURE MSL

ANALYTICAL NOTE

The narrative account describes conditions as Beaufort force 9 to 10 (47–55 knots sustained). ERA5 reconstruction shows 10 m sustained wind of 27.0 knots (Beaufort force 6–7) at the anchorage position. These are not equivalent measurements and the difference is analytically significant.

ERA5 reports the 10-metre neutral-stability wind speed averaged over a ~31 km grid cell. The MAIB Beaufort classification is derived from sea-state observation at the vessel's position — a method that conflates locally generated wind waves with remotely generated swell.

The dominant forcing mechanisms for anchor drag in this case were heavy swell loading on a vessel in ballast combined with severe gust loading (45 kt at 10 m, likely exceeding 50 kt at bridge height). Sustained local Force 9–10 wind was not required to produce the observed outcome.

DIVERGENCE CLASSIFICATION

PARAMETER	NARRATIVE	RECONSTRUCTION	CLASSIFICATION
Wind speed (sustained)	47-55 kt	27.0 kt	OVERSTATED
Significant wave height	4-5 m	4.1 m	CONFIRMED

Unit Conversion Chain — Wind Speed

ERA5 source: **13.875** m/s → × (3600 / 1852) = **27.0** kt

Unit Conversion Chain — Pressure

ERA5 source: **100438.8** Pa → ÷ 100 = **1004.4** hPa

Unit Conversion Chain — Temperature

ERA5 source: **284.956** K → − 273.15 = **11.8** °C

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Decision-Time Environment

The following chronology presents events with environmental context at each decision gate.

Day 1 — 1700 UTC

Vessel anchored in the bay

Port anchor deployed with 7 shackles of cable in 17 m charted depth, approximately 1 nm from the beach.

Day 3 — Evening

Decision: remain at anchor overnight

Master aware of forecast: southerly winds up to Beaufort force 8 (34–40 kt). Night orders issued: engine readiness required at wind speeds >16–17 m/s.

```
Forecast: S'ly F8 (34–40 kt) | Decision window: OPEN
```

Day 3 — 2100 UTC

First threshold exceedance (ERA5)

All 3 environmental signals (wind, gust, wave) exceed safety thresholds simultaneously. Intervention window opens.

```
20.1 kt sustained, 32.0 kt gusts, 2.7 m SWH | Risk: 60.1% ELEVATED
```

Day 4 — 0000 UTC

Storm escalation

Environmental conditions escalate to EXTREME risk band.

```
23.5 kt sustained, 37.9 kt gusts, 3.3 m SWH | Risk: 91.4% EXTREME
```

Day 4 — 0400 UTC

Chief officer takes bridge watch

Southerly wind gusting over 52 knots. Swell 4–5 metres. Vessel yawing at anchor.

```
26.2 kt sustained, 43.7 kt gusts, 4.0 m SWH | Risk: 100% EXTREME
```

Day 4 — 0425 UTC

Anchor starts to drag

Anchor watch alarm sounds. Master called, engine room alerted.

```
Decision window: CLOSING
```

Day 4 — ~0500 UTC

Anchor aweigh — fouled by chain

Anchor fouled by substantial length of chain. Vessel fell beam-to the wind.

```
Decision window: CLOSED
```

Day 4 — 0525 UTC

Grounding

Vessel grounded on the beach.

```
27.0 kt sustained, 44.8 kt gusts, 4.1 m SWH | Risk: 100% EXTREME
```

Day 4 — 1415 UTC

Refloated

Vessel refloated by local harbour tugs and own main engine.

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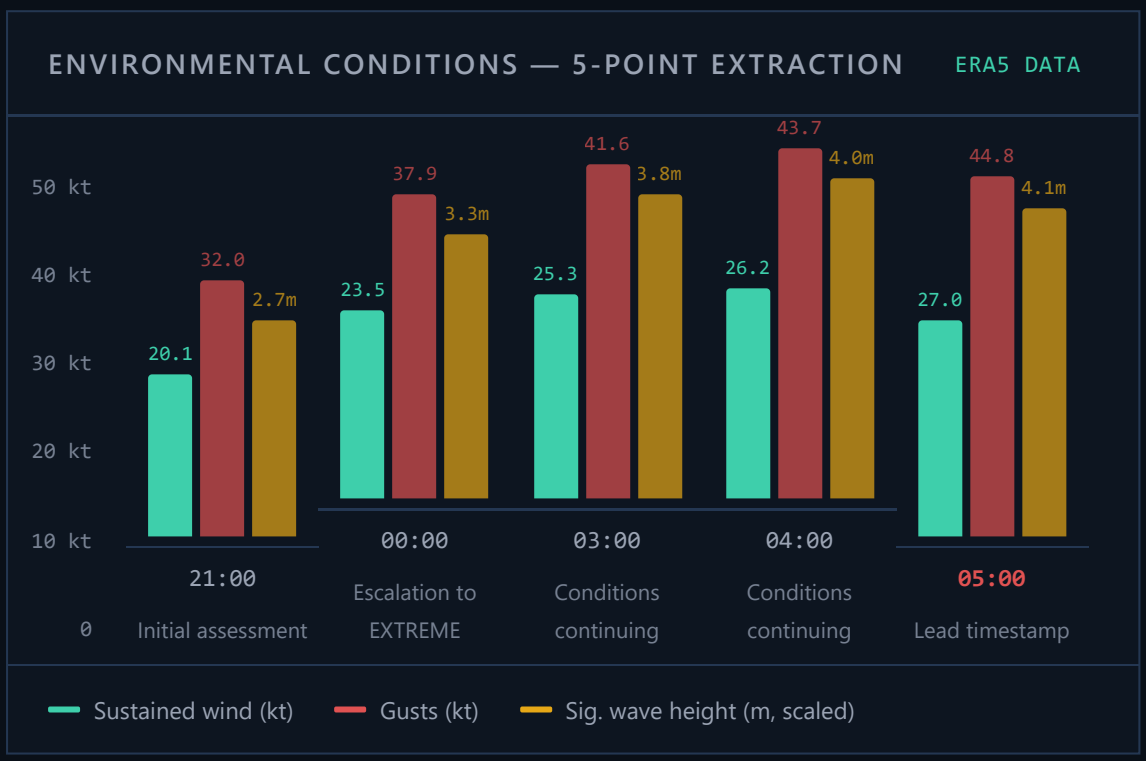
Environmental Escalation — Verified Profile

Verified environmental escalation profile across the incident window.



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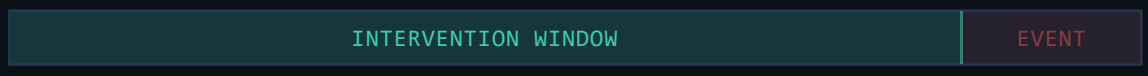
Threshold Exceedance Analysis

Threshold exceedance detection identifies the first point at which environmental conditions exceeded safety thresholds — determining the intervention window available before the incident.

Threshold Exceedance Findings

FIRST EXCEEDANCE	2018-12-17 21:00 UTC
EVENT TIME	05:00 UTC
INTERVENTION WINDOW	480 minutes (8.0 hours)
SIGNALS EXCEEDED	Gust speed, Significant wave height, Sustained wind
STRENGTH AT DETECTION	WEAK
STRENGTH PROGRESSION	WEAK → WEAK → MODERATE → MODERATE → MODERATE



21:00 UTC — First threshold exceedance

05:00 UTC — Incident

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Hazard Acceleration Profile

Classification of the rate of environmental deterioration across the incident window.

Deterioration Classification

COMPOSITE	DETERIORATING / GRADUAL
CHANNELS WORSENING	3 simultaneously
DOMINANT CHANNEL	Sustained wind
ONSET CLASSIFICATION	sustained and predictable deterioration

CHANNEL	TREND	RATE
Sustained wind	DETERIORATING	GRADUAL
Gust speed	DETERIORATING	GRADUAL
Significant wave height	DETERIORATING	GRADUAL

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Time to Irreversibility

Regime Classification: SAFE

T_irr >> T_practical. Wide recovery margin. Intervention was physically possible throughout the incident window.

T_IRR (CONSERVATIVE)	**16.5 minutes**
T_IRR (BEST ESTIMATE)	20.6 minutes
T_PRACTICAL	6.0 minutes
MARGIN (CONSERVATIVE)	**+10.5 minutes**
MARGIN (BEST ESTIMATE)	+14.6 minutes
BINDING CONSTRAINT	Spatial drift — distance from anchored position to lee shore

MECHANISM & INFORMATION

FAILURE MECHANISM	Gradual degradation
MECHANISM CONFIDENCE	HIGH

DESCRIPTION

Environmental conditions deteriorated progressively, eroding safety margins incrementally. Primary mechanism matched 4 of 5 diagnostic features. Secondary mechanism (spatial phase transition) also matched, reflecting the anchor drag as a spatial boundary crossing.

INFORMATION CHANNELS	3/3 — SUFFICIENT

T_IRR INTERPRETATION

T_irr estimates the physical time before the vessel's situation becomes irreversible. T_practical estimates the minimum time a competent crew would need to recognise the hazard, assess the situation, and begin effective intervention. The margin is the difference: positive margin means intervention was physically possible.

Drift rate not stated in the MAIB report. A conservative default of 1.5 m/s was applied from Admiralty Sailing Directions (published upper-bound tidal stream rate for UK coastal waters without engine power). The actual drift rate during anchor drag (approximately 1 nm in 60 minutes, with leeway increasing to over 2 knots as the vessel fell beam-to the wind) was slower than this default, meaning the true T_irr was likely longer than stated here.

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Risk Analysis

100% Risk Band: EXTREME

Composite environmental risk score at lead timestamp

RISK DRIVER	VALUE
Sustained wind	27.0 kt
Gust speed	44.8 kt
Significant wave height	4.1 m
Wind vs tide	with
Visibility	moderate
Wind shear (10-100m)	12.7 kt
Dewpoint spread	3.2 °C

MULTIPLIER	FACTOR	STATUS	RATIONALE
Proximity	x1.20	ACTIVE	*Proximity factor: vessel operating within 1 nm of shore.*
Circadian	x1.70	ACTIVE	*Fatigue-adjusted: incident occurred during hours of reduced alertness (0300–0600 UTC).*

MULTIPLIER	FACTOR	STATUS	RATIONALE
Cold water	x1.50	ACTIVE	*Environmental severity factor: sea surface temperature below cold-water threshold.*

◆ FORESEEABILITY TEST

This report assesses foreseeability by applying the following test:

- The environmental data required to assess the risk existed and was accessible at the time of the decision (ERA5 reanalysis confirms this retrospectively; forecast data confirms it prospectively).
- The data was sufficient to identify that conditions at the anchorage position would exceed the operational envelope for a vessel of this type, size, and ballast condition.
- A reasonable operator with access to this data would have identified the risk of anchor drag under the forecast conditions and considered proceeding to sea or relocating to a more sheltered anchorage.
- The environmental progression from forecast conditions to actual conditions was continuous, not discontinuous — there was no sudden, unforeseeable step-change that made the outcome unavoidable.

This report does not determine causes or assign responsibility. The foreseeability test evaluates environmental evidence only.

Key Analytical Finding

KEY FINDING

Environmental reconstruction separates local wind forcing from remotely generated swell, revealing that the conventional Beaufort force 9–10 classification in the narrative account conflated two distinct physical mechanisms. ERA5 reanalysis shows 10 m sustained winds of 20.1–27.0 knots (Beaufort force 5–7) at the anchorage position, with gusts of 32.0–44.8 knots and significant wave height of 2.7–4.1 metres. All three environmental signals exceeded safety thresholds at 21:00 UTC, providing an 8-hour intervention window before the grounding.

PIPELINE CLASSIFICATION

CRITICAL: Environmental reconstruction shows the vessel entered an ELEVATED risk regime at 21:00 UTC and an EXTREME risk regime by 00:00 UTC — five full hours before the grounding. The dominant forcing was heavy swell (3.8–4.1 m SWH) propagating into the anchorage from the open English Channel, combined with severe gust loading (41.6–44.8 kt). Sustained 10 m wind of 20–27 kt (Beaufort force 5–7) was materially lower than the narrative Force 9–10 description.

This finding is derived from the quantified environmental reconstruction, not from the narrative account. It represents what was objectively knowable at the time the critical operational decision was made, based on environmental data that existed and was accessible.

This report is an environmental reconstruction. It does not make determinations about individual conduct, operational adequacy, or regulatory compliance. Its purpose is to quantify the environmental conditions under which operational decisions were made.

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Questions This Report Does Not Answer

The following questions are outside the scope of this reconstruction:

- Whether the master's decision to remain at anchor was reasonable given the information available to him at the time.
- Whether the vessel's anchor and cable were in adequate condition for the expected storm.
- Whether alternative anchorages or sea room were available and practicable given the vessel's fuel and operational constraints.
- Whether the fouled anchor chain was a pre-existing hazard that should have been identified.
- Whether the port authority or vessel operator should have intervened before the storm arrived.

Analytical Boundaries

The following boundaries apply to this reconstruction. Where a boundary introduces directional bias, the direction is stated explicitly so that the reader can assess the weight to place on each finding.

- ERA5 reconstructs conditions on a ~31 km grid. It does not resolve harbour-scale wind acceleration, cliff-edge effects, or shallow-water wave transformation at the anchorage. Where these effects are present, actual conditions at the vessel's position are typically more severe than the gridded values. The environmental values in this report should therefore be interpreted as conservative lower bounds for the conditions experienced at the anchorage.
- ERA5 gust speed represents the maximum instantaneous wind at the 10-metre reference height. On a 180-metre vessel in ballast with high windage, the wind speed at bridge height (~25–30 m) would exceed the 10-metre value due to the vertical wind shear profile. The gust values presented in this report understate the actual wind loading on the vessel's superstructure.
- This reconstruction quantifies the environmental conditions surrounding the incident. Vessel-specific margin analysis — such as anchor holding capacity versus environmental loading — was outside scope. The environmental reconstruction is complete; the vessel-response modelling was not attempted for this case.
- ERA5 significant wave height represents open-ocean conditions. In a coastal anchorage, shoaling, refraction, and harbour geometry can modify wave characteristics. Depending on bathymetry and exposure, these effects may increase or decrease wave loading at the vessel's position relative to the open-ocean value. The ERA5 wave height of 4.1 m aligns closely with the 4–5 m range described in the narrative account, suggesting the open-ocean value is representative for this location.
- ERA5 provides data at standard hourly intervals. The anchor drag began at 04:25 UTC, between the 04:00 and 05:00 extraction points. Both extraction points classify as EXTREME risk (100%). The environmental state at 04:25 falls within the range bounded by these two verified data points.

Evidence Integrity

All environmental values are traceable to source data through an auditable processing chain.

PIPELINE VERIFICATION

DATA SOURCE	ECMWF ERA5 Atmospheric Reanalysis
ERA5 RESOLUTION	~31 km horizontal, hourly
PROCESSOR VERSION	5.4.3
RUNNER VERSION	2.4.3
GATE CONFIG HASH	989aadf7b9b1a4acef8606e162015337e75a8ce5fcb4edfb1e0d79f4d59513aa
RECONSTRUCTION TIMESTAMP	2026-04-12T16:30:31Z
CASE POSITION	50.15°N, -5.07°W

EVIDENCE ARTIFACT	HASH
Input file (fig_inputs.json)	716c5d1bb3a3c9be920e07cb2ca2db9dda1f1b028c1ce9ef96a2b44913fb16b8
Batch file	1b3cf1b44dd5a165a5d11ae8bf2dfab8a988bd952640255f773b2f04d3754396
Risk assessment policy	e55d9cb6149637be771860a6aa5d8a7213f181c7a8366ac55b905ee6ea69fe3b

EVIDENCE ARTIFACT	HASH
Unit gate config	989aadf7b9b1a4acef8606e162015337e75a8ce5fcb4edfb1e0d79f4d59513aa
Threshold detection config	b00e295ff4061814
Threshold detection analysis	1ad055ab2957b83595a5af7dd3856b761f524421c628179d1a71c87c4eed72d0
Hazard acceleration config	7961f7bf72ed3941
Hazard acceleration analysis	778a82d208a402b164c3e0ce73c24739761522e4a1caadb312c7a201bf5f6561
Risk input (21:00)	09ed30171d8bf50c72831cf0f9d0aa6b0d3c0ef22133eb7d9ea715980381056f
Risk analysis (21:00)	e9c26b358fe007b20f25b034f1d21e70e9e3b5af406d57e5edadd454d73b4f86
Risk audit (21:00)	753242f3bfc62fb95b60b358a6187a78a9df06d8698e900cf34cb2963c3be6cb
Risk input (00:00)	3c0389b8a74cd14fd21d8d60824fa28abd3c21df8cb102f87c8b01fe57ae61ce
Risk analysis (00:00)	e11290148028b154eb5a51117f5da015ddf376ff4e88629083cba35b0cd78570
Risk audit (00:00)	c1ebed544d710ce2292b4a3a1411be90f28cce5accfb3bfb36dc0478a950438e
Risk input (03:00)	572924c5b68e7178cab9ef55ba98dd8eec195e23d85ec1403057e81c5baeb300
Risk analysis (03:00)	57b092446fe636bee6a6d27bea49b47d480d067d90934e8fbf1928ca93b77c61
Risk audit (03:00)	d71c86918e2ca2c63dd32cbe55b82cefbc06949bb8e6ec3448e3ffcd7ee8f257

EVIDENCE ARTIFACT	HASH
Risk input (04:00)	177315afaad6446444160d2493f3215a5b4fa793e2e033350e6f123c07a6795d
Risk analysis (04:00)	203be4ebfe8871538e057f0f85521f5828cc0d8114c3195ae8f195da56dad8a0
Risk audit (04:00)	8580c90510ce739180fc1673abf0cda25872803814021d9672202e8e6b4f9f16
Risk input (05:00)	d60f707c008726574c6fca03e929fd98666e78be085a3a7b49a12a7225f8b928
Risk analysis (05:00)	de71ddc622ed71c22b68cbe0ff6e0f17e73985f42016df9e3ae6d83d94f85f78
Risk audit (05:00)	32b49d0c134bc49355b6dcbca75b8711f4b5c3569cabf0345e3c384004828750

REPRODUCIBILITY

Any party with access to the same ERA5 data, the published FishIntel pipeline version, and the gate configuration hash can independently reproduce the environmental values in this report. The reconstruction is deterministic: identical inputs produce identical outputs.

Methodology

FishIntel Global reconstructs environmental conditions using ECMWF ERA5 atmospheric reanalysis — the same dataset relied upon by meteorological agencies, climate researchers, and aviation safety authorities worldwide.

ERA5 provides hourly estimates on a global grid at approximately 31 km horizontal resolution. All values pass through a unit-gated validation layer enforcing explicit conversion factors and plausibility bounds using Decimal arithmetic. The pipeline is deterministic: identical inputs always produce identical outputs.

This reconstruction operates within defined boundaries. ERA5 does not capture localised effects smaller than its grid resolution. Where a case involves mesoscale phenomena, the reconstruction presents ERA5 as the synoptic baseline, not the peak event conditions.

FishIntel Analytical Architecture

FishIntel operates as a multi-system analytical architecture designed to reconstruct the environmental conditions surrounding maritime incidents and quantify the operational margin before irreversible outcomes occur. The architecture consists of several interlocking analytical systems:

ARCHITECTURE COMPONENTS

SOURCE EXTRACTION SYSTEM

Converts investigation material and case information into structured machine-readable inputs.

ENVIRONMENTAL RECONSTRUCTION SYSTEM

Retrieves and reconstructs decision-time environmental conditions using ERA5 atmospheric reanalysis.

UNIT INTEGRITY SYSTEM

Validates and canonicalises all physical measurements using explicit conversion factors and Decimal arithmetic.

RISK INFERENCE SYSTEM

Evaluates reconstructed environmental conditions to produce structured risk assessments.

THRESHOLD DETECTION SYSTEM

Identifies the first point at which environmental conditions exceed safety thresholds and quantifies the intervention window.

HAZARD ACCELERATION SYSTEM

Classifies the rate and pattern of environmental deterioration across the incident window.

EVIDENCE CHAIN SYSTEM

Generates deterministic seals, hashes, and verification artifacts for every analytical step.

REPORT GENERATION SYSTEM

Renders validated outputs into investigation-grade analytical reports.

These systems operate together within a deterministic reconstruction pipeline. Each system produces independently verifiable outputs with cryptographic provenance.

ANALYTICAL INDEPENDENCE

FishIntel Global is an independent environmental reconstruction service. This report was produced without instruction from any party to the incident, its investigation, or any related proceedings. FishIntel's role is complementary to national maritime accident investigation.

Closing Statement

Maritime incidents are frequently described after the fact using qualitative or semi-quantitative environmental descriptions: "strong winds," "heavy swell," "poor visibility." These descriptions are valuable for narrative understanding but do not quantify the decision-time environment with the precision required by insurers, regulators, and legal professionals.

Environmental reconstruction bridges this gap. By reconstructing the conditions that existed at the time a critical decision was made — using data sources that were available or forecastable at that time — it becomes possible to evaluate what was objectively knowable before the outcome occurred.

This does not mean that every incident was preventable. It means that the environmental context of every maritime decision can now be quantified, compared against published forecasts, and assessed for foreseeability. The data exists. The question is whether it was used.

FishIntel Global — Maritime Foreseeability Intelligence
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